Computer fundamentals

How a computer takes input, processes data and produces useful output.

Laboratory 09–10
An educational presentation in LibreOffice Impress

The input–process–output cycle

INPUT		PROCESS		OUTPUT
Keyboard Mouse Microphone	→	CPU follows program instructions	→	Screen Printer Speakers

Example: type 2 + 3, let the CPU calculate, then read 5 on the screen.

Hardware and software

HARDWARE

Physical parts you can touch

Processor, keyboard and screen

SOFTWARE

Instructions that hardware runs

Operating system and applications

Memory and storage

RAM

Holds data while programs run.

Loses its contents when power is removed.

SSD / HARD DRIVE

Keeps saved files and programs.

Retains data when the computer is switched off.

The operating system

The operating system manages hardware and running programs.

It provides tools to organise files and folders.

Applications help us complete tasks, such as writing a report or calculating a marksheet.

Multimedia: sound as output

A computer can represent sound as digital samples.

Speakers turn an electrical signal into audible vibrations.

This slide includes a short, original three-note audio clip.

The clip plays on entry. A separate WAV file is also included.

Quick check

1. Is a microphone an input device or an output device?

2. Which keeps saved files after shutdown: RAM or an SSD?

3. Name one operating system and one application.

Answers: input device; SSD; for example, Linux and LibreOffice.